

Project 3: Active Learning for Learning-to-Defer

Human-Centric Artificial Intelligence

Chidambaram Muthukumar – 641326

1. Introduction

This report presents the design and evaluation of a human-AI collaborative system for news topic classification, built on the AG News dataset. The system combines a trained classifier with a simulated human expert through a learning-to-defer mechanism, and extends this to a setting where expert labels are not available during training, requiring active learning to efficiently discover when deferral is beneficial.

Four tasks are addressed in sequence: (1) training a baseline classifier on the full labeled dataset, (2) designing and analyzing a simulated expert with realistic, non-uniform competence, (3) implementing a learning-to-defer strategy that combines classifier and expert predictions, and (4) implementing an active learning strategy to discover expert competence without upfront access to expert labels.

2. Task 1: Baseline Classifier

2.1 Model choice

The classification model chosen for this task is a TF-IDF vectorizer combined with a multinomial Logistic Regression classifier. This choice was made for tractability: fine-tuning a transformer-based model such as BERT on 120,000 training articles would require substantially more compute and training time than is practical for this course project, while a TF-IDF and linear model pipeline trains in under a minute on a standard laptop CPU and is fully interpretable. We consider a transformer-based approach a natural extension of this work, discussed further in the conclusion.

The classifier was trained on the complete AG News training set (120,000 articles, 4 balanced classes: World, Sports, Business, Sci/Tech) using a TF-IDF representation restricted to the 20,000 most informative terms, with English stop words removed.

2.2 Results

Metric	Value
Model	TF-IDF + Logistic Regression
Training set size	120,000 articles
Test set size	7,600 articles
Test accuracy	91.53%

This baseline accuracy of 91.53% serves as the reference point that the human-AI collaborative system (Tasks 3 and 4) should aim to match or surpass.

3. Task 2: Simulated Expert

3.1 Expert design

The simulated expert was designed to be deliberately imperfect, with competence that varies by topic rather than being uniform across the input space, as required by the task. Specifically:

- On World and Sports articles, the expert is correct 95% of the time, reflecting strong, reliable expertise in these topics.
- On Business and Sci/Tech articles, the expert is correct only 60% of the time. When wrong, the expert systematically confuses Business with Sci/Tech and vice versa, rather than making arbitrary errors.

This design is realistic: Business and Sci/Tech news articles frequently share vocabulary (companies, technology investment, markets, product launches), making them genuinely difficult to distinguish, even for a competent human reader. World and Sports articles, by contrast, tend to use more topic-specific and unambiguous vocabulary.

3.2 Strengths and weaknesses analysis

The expert was evaluated against the ground-truth labels of the full AG News test set (7,600 articles). Results are reported overall and broken down per class.

Class	Expert accuracy
World	95.11%
Sports	95.42%
Business	59.79%
Sci/Tech	60.74%
Overall	77.76%

The confusion matrix below confirms the intended competence profile: World and Sports show strong diagonal dominance, while Business and Sci/Tech show a large, near-symmetric block of confusion between each other (764 Business articles misclassified as Sci/Tech, 746 Sci/Tech articles misclassified as Business), with almost no confusion crossing into World or Sports.

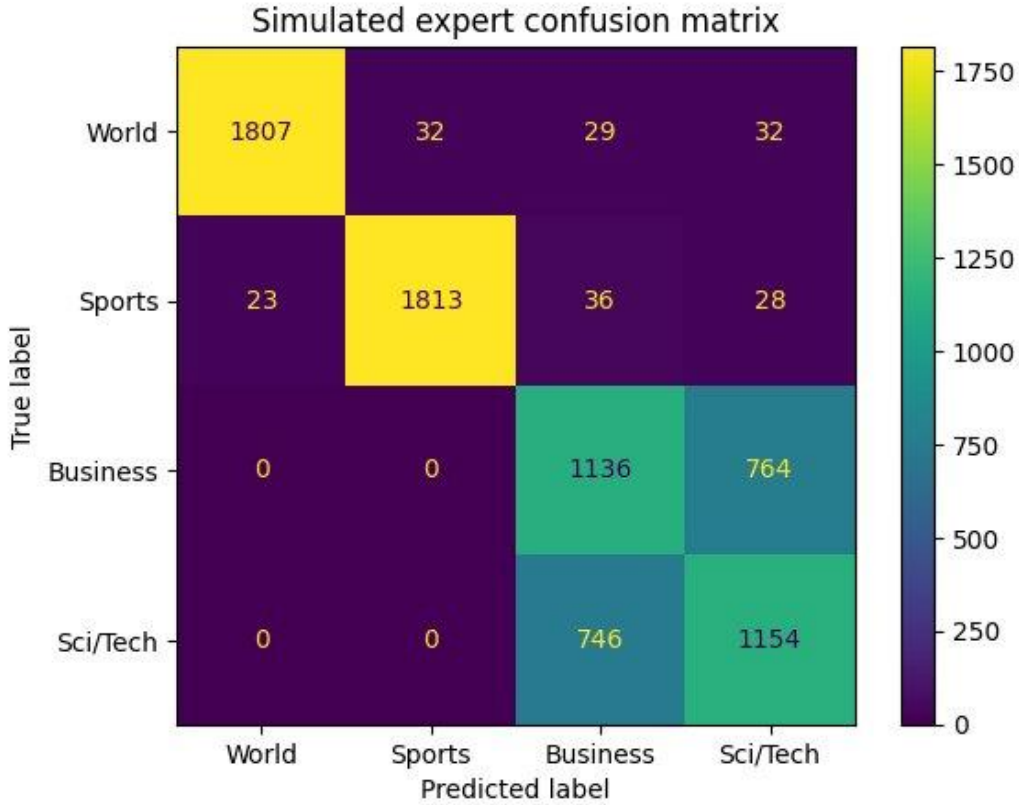


Figure 1. Confusion matrix of the simulated expert on the AG News test set.

4. Task 3: Learning to Defer

4.1 Method

The learning-to-defer strategy implemented is confidence-based deferral. For each test example, the classifier produces a predicted probability distribution over the four classes. If the maximum predicted probability falls below a threshold τ , the system defers the decision to the expert; otherwise, it trusts the classifier's own prediction. This approach was chosen for its simplicity, interpretability, and direct connection to model calibration: low-confidence predictions are, on average, more likely to be wrong, making them good candidates for deferral.

4.2 Results

The threshold τ was swept across a range of values to characterize the trade-off between deferral rate and system accuracy.

τ	System acc.	Defer rate	Clf. acc. (deferred)	Expert acc. (deferred)
0.30	91.54%	0.03%	50.00%	100.00%
0.40	91.86%	1.04%	40.51%	72.15%
0.50	92.25%	3.80%	52.60%	71.63%

0.60	92.59%	8.96%	59.32%	71.22%
0.70	92.25%	14.66%	66.79%	71.72%
0.80	91.03%	23.18%	74.69%	72.53%
0.90	88.74%	38.51%	81.79%	74.55%

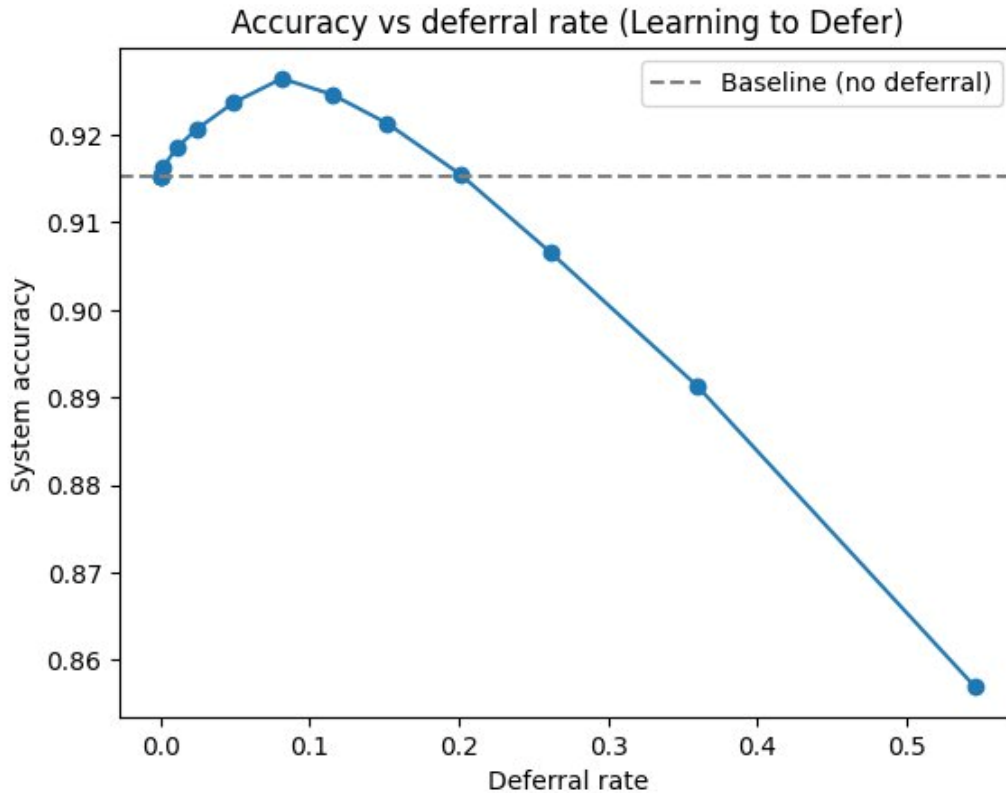


Figure 2. System accuracy as a function of deferral rate, compared to the AI-only baseline.

4.3 Discussion

The accuracy-versus-deferral-rate curve follows an inverted-U shape. At low thresholds, almost no examples are deferred, so system accuracy remains close to the 91.53% baseline. As the threshold increases, accuracy rises, peaking at 92.59% when $\tau = 0.6$, an absolute improvement of 1.06 percentage points over the AI-only baseline.

Beyond this point, accuracy declines. The reason is visible in the last two columns of the results table: as the threshold increases further, the classifier's own accuracy on the deferred cases rises substantially (from 50% at $\tau = 0.3$ to 81.79% at $\tau = 0.9$), meaning the system is increasingly deferring cases the classifier could have answered correctly itself, to an expert whose accuracy on those same cases plateaus around 71–75%. Past the optimum, deferral substitutes good classifier predictions with a less reliable expert opinion, which is detrimental to overall accuracy. This crossover illustrates that the value of deferral depends entirely on whether the party receiving the decision is actually better suited to it, not merely on how confident the classifier is.

5. Task 4: Active Learning for Expert Competence Discovery

5.1 Setting and strategy

In this task, no expert labels are available during training. The deferral threshold must instead be learned by actively querying the expert on a limited number of selected examples drawn from the training pool, and using only those queried labels to estimate a good threshold.

The active learning strategy adopted is uncertainty sampling: in each round, the examples with the lowest maximum predicted probability (i.e., the classifier's least confident predictions) are selected for querying. This strategy was chosen because it directly targets the region of the input space where the deferral decision actually matters — querying the expert on examples the classifier is already confident about provides little information about when to defer, since deferral would rarely be triggered there in the first place.

As a baseline for comparison, an identical loop using uniformly random sampling of query examples was also implemented. In both cases, after each round of queries, the threshold that maximizes accuracy on the queried examples so far is selected and then evaluated on the held-out test set (using the test-set expert simulation only for evaluation, never for threshold selection).

5.2 Results

Queries	Uncertainty: threshold	Uncertainty: acc.	Random: acc.
50	0.45	92.03%	91.54%
100	0.50	92.25%	91.54%
150	0.50	92.25%	92.25%
200	0.55	92.53%	91.54%
250	0.55	92.53%	91.86%
300	0.55	92.53%	91.86%
350	0.60	92.59%	92.25%
400	0.55	92.53%	92.25%
450	0.55	92.53%	91.86%
500	0.55	92.53%	91.86%

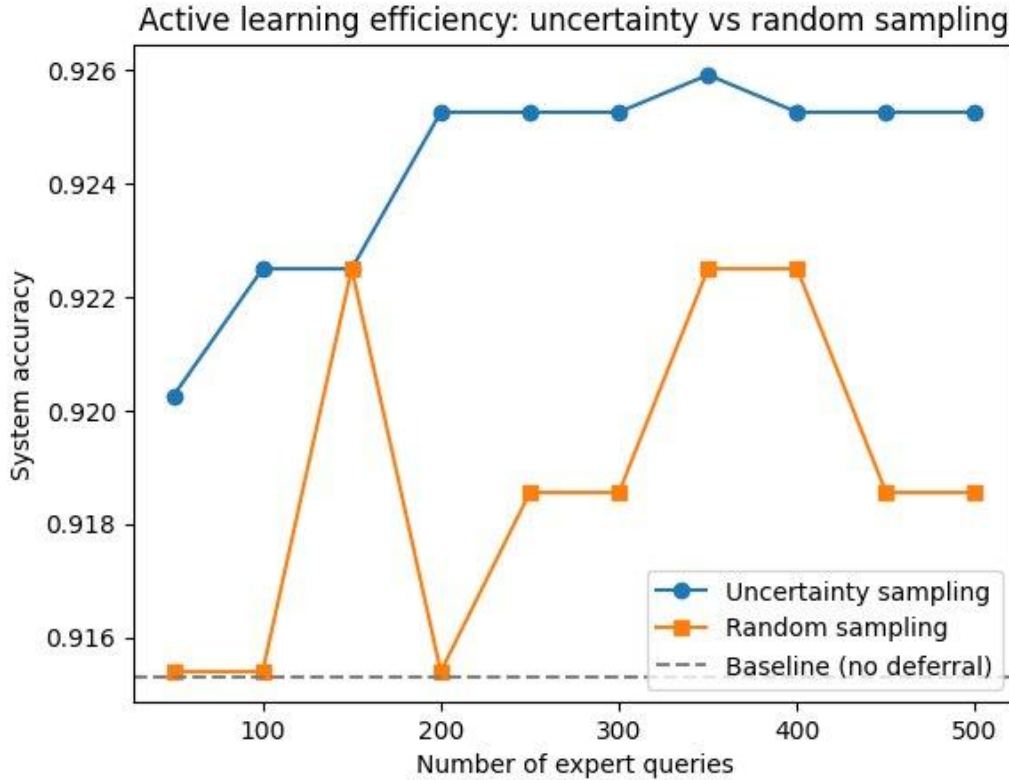


Figure 3. System accuracy versus number of expert queries, comparing uncertainty sampling and random sampling.

5.3 Discussion

Uncertainty sampling converges quickly and stably: by round 3 (200 queries), it has already identified a threshold of 0.55, very close to the 0.6 optimum found via full grid search with complete information in Task 3, and the learned threshold remains stable for the remainder of the experiment. The resulting system accuracy of 92.53% is within 0.06 percentage points of the best result obtainable with full knowledge.

Random sampling, by contrast, is markedly less efficient. Its threshold estimate fluctuates between 0.3 and 0.7 across rounds even after 500 queries, never stabilizing, and its best system accuracy across all rounds reaches only 92.25%, with most rounds sitting closer to 91.5–91.9%, barely above the no-deferral baseline.

This gap arises because uncertainty sampling concentrates its limited query budget precisely in the region of the input space where the deferral decision is actually exercised — low-confidence predictions — while random sampling wastes most of its queries on high-confidence examples where the classifier is already correct and the choice of threshold is irrelevant. As a result, uncertainty sampling reaches a near-optimal deferral policy using a fraction of the labeled examples that random sampling would require to achieve comparable stability.

6. Conclusion

This project implemented and evaluated a human-AI collaborative system for AG News topic classification across four stages. A TF-IDF and Logistic Regression baseline classifier reached

91.53% test accuracy. A simulated expert with realistic, non-uniform competence (strong on World/Sports, weak and systematically confused between Business/Sci-Tech) was designed and analyzed, reaching 77.76% overall accuracy. A confidence-based learning-to-defer system combining the two improved system accuracy to 92.59% at an optimal threshold of 0.6, demonstrating that deferral is beneficial only up to the point where the expert remains more reliable than the classifier on the specific cases being deferred.

Finally, an active learning strategy based on uncertainty sampling was shown to discover a near-optimal deferral threshold using only a few hundred expert queries, substantially outperforming random query selection in both convergence speed and stability. This confirms that targeting the classifier's least confident predictions is an effective and efficient strategy for learning expert competence profiles without requiring expert labels during initial training.

A natural extension of this work, suggested as future work, is to replace the TF-IDF and Logistic Regression baseline with a fine-tuned transformer model, and to implement the optional interactive interface (Task 5) allowing a real human to act as the expert rather than relying on simulation.